Linear Algebra with Applications in Automatic Control

School of Automation,
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Fall Semester

3.4 Minimal polynomials and the first decomposition of a linear space

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•3.4.1 Annihilating and minimal polynomials-definition

Let $\mathcal{V}$ be linear space of dimension n over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, then for any nonzero vector $\boldsymbol{\alpha} \in \mathcal{V}$ a sequence of vectors can be generated

$$\boldsymbol{\alpha}, \quad \mathscr{A}(\boldsymbol{\alpha}), \quad \mathscr{A}^2(\boldsymbol{\alpha}), \quad \ldots, \quad \mathscr{A}^m(\boldsymbol{\alpha}), \quad \ldots \quad (1)$$

Since all the vectors in (1) are in the n-dimensional space $\mathcal{V}$, when m is large enough, the $m+1$ vectors

$$\boldsymbol{\alpha}, \quad \mathscr{A}(\boldsymbol{\alpha}), \quad \mathscr{A}^2(\boldsymbol{\alpha}), \quad \ldots, \quad \mathscr{A}^m(\boldsymbol{\alpha})$$

must be linearly dependent, especially, when $m = n$, there will be $n+1$ vectors and they are always linearly dependent.

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•3.4.1 Annihilating and minimal polynomials -definition

Thus, there exist a_0, a_1, ..., a_m, not all zeros, such that the linear combination of the above vectors are zero, that is

$$\mathbf{0} = a_0\boldsymbol{\alpha}+a_1\mathscr{A}(\boldsymbol{\alpha})+a_2\mathscr{A}^2(\boldsymbol{\alpha})+\cdots+a_m\mathscr{A}^m(\boldsymbol{\alpha}) \triangleq \varphi_{\boldsymbol{\alpha}}(\mathscr{A})(\boldsymbol{\alpha})$$

where

$$\varphi_{\boldsymbol{\alpha}}(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + a_m\lambda^m \in \mathcal{F}[\lambda]\,,$$

-see example 2.8 for the definition of a polynomial of a linear transformation.

Hence, for any $\boldsymbol{\alpha} \in \mathcal{V}$ there exists $\varphi_{\boldsymbol{\alpha}}(\lambda) \in \mathcal{F}[\lambda]$ such that $\varphi_{\boldsymbol{\alpha}}(\mathscr{A})(\boldsymbol{\alpha}) = \mathbf{0}$. From this observation we introduce the following concept.

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•3.4.1 Annihilating and minimal polynomials-definition

Definition 3.13 Let $\mathcal{V}$ be n-dimensional linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. The given polynomial $\varphi_{\boldsymbol{\alpha}}(\lambda) \in \mathcal{F}[\lambda]$ is called an annihilating polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$ if

$$\varphi_{\boldsymbol{\alpha}}(\mathscr{A})(\boldsymbol{\alpha}) = \mathbf{0}, \tag{1}$$

where $\varphi_{\boldsymbol{\alpha}}(\mathscr{A}) = a_0\mathscr{E} + a_1\mathscr{A} + a_2\mathscr{A}^2 + \cdots + a_m\mathscr{A}^m$. $\psi_{\boldsymbol{\alpha}}(\lambda)$ is called the minimal polynomials of $\mathscr{A}$ at $\boldsymbol{\alpha}$ if it is an annihilating polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$ and a divisor of any other annihilating polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$.

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•3.4.1 Annihilating and minimal polynomials-properties

Theorem 3.19 Let $\mathcal{V}$ be an n-dimensional linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\alpha_{\boldsymbol{\alpha}}(\lambda)$ be the minimal polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$.

1. If $\varphi_{\boldsymbol{\alpha}}(\lambda)$ is an annihilating polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$, then $\alpha_{\boldsymbol{\alpha}}(\lambda)|\varphi_{\boldsymbol{\alpha}}(\lambda)$;

2. if $\beta_{\boldsymbol{\alpha}}(\lambda)$ is a minimal polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$, then $\beta_{\boldsymbol{\alpha}}(\lambda) = \alpha(\lambda)$.

Corollary 3.3 Let $\mathcal{V}$ be n-dimensional linear space over $\mathcal{F}$, $\alpha_{\boldsymbol{\alpha}}(\lambda)$ be the minimal polynomial of $\mathscr{A} \in \mathcal{L}(\mathcal{V})$ at $\boldsymbol{\alpha}$. Then for any $\beta(\lambda) \in \mathcal{F}[\lambda]$ with $\deg \beta(\lambda) < \deg \alpha_{\boldsymbol{\alpha}}(\lambda)$, $\beta(\mathscr{A})(\boldsymbol{\alpha}) \neq \mathbf{0}$.

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- Annihilating and minimal polynomials of a linear transformation $\mathscr{A}$ at a (sub)space

Definition 3.14 Let $\mathcal{V}$ be n-dimensional space over $\mathcal{F}$, $\mathcal{W} \subset \mathcal{V}$ be a subspace, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$. $\psi_{\mathcal{W}}(\lambda) \in \mathcal{F}[\lambda]$ is an annihilating polynomial of $\mathscr{A}$ at $\mathcal{W}$ if

$$\psi_{\mathcal{W}}(\mathscr{A})(\boldsymbol{\alpha}) = \mathbf{0} \quad \forall \boldsymbol{\alpha} \in \mathcal{W}. \tag{1}$$

The monic annihilating polynomial of $\mathscr{A}$ at $\mathcal{W}$ with the lowest degree is called the minimal polynomial of $\mathscr{A}$ at $\mathcal{W}$.

Let $\mathcal{W} = \mathcal{V}$. Then annihilating polynomials and the minimal polynomial of $\mathscr{A}$ at the whole space $\mathcal{V}$ can be defined. For brevity they are called annihilating polynomials and the minimal polynomial of $\mathscr{A}$ respectively.

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- Annihilating and minimal polynomials of a linear transformation $\mathscr{A}$ at a subspace--properties

•Remark

Let $\dim \mathcal{W} = m$, $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_m$ be a basis of $\mathcal{W}$, then $\psi_{\mathcal{W}}(\lambda)$ is an annihilating polynomial of $\mathscr{A}$ at $\mathcal{W}$, if and only if

$$\psi_{\mathcal{W}}(\mathscr{A})([\boldsymbol{\beta}_1 \ \ \boldsymbol{\beta}_2 \ \ \ldots \ \ \boldsymbol{\beta}_m]) = [\underbrace{\mathbf{0} \ \ \mathbf{0} \ \ \ldots \ \ \mathbf{0}}_{m}],$$

that is, $\psi_{\mathcal{W}}(\lambda)$ is annihilating polynomial at each of the basis vectors $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_m$.

Let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ be a basis of $\mathcal{V}$, then $\varphi(\lambda) \in \mathcal{F}[\lambda]$ is an annihilating polynomial of $\mathscr{A}$, if and only if

$$\varphi(\mathscr{A})([\boldsymbol{\alpha}_1 \ \ \boldsymbol{\alpha}_2 \ \ \ldots \ \ \boldsymbol{\alpha}_n]) = [\underbrace{\mathbf{0} \ \ \mathbf{0} \ \ \ldots \ \ \mathbf{0}}_{n}]. \quad (1)$$

Since the vectors $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ are linearly independent, (1) holds if and only if $\varphi(\mathscr{A})$ is the zero transformation, that is $\varphi(\mathscr{A}) = \mathscr{O}$.

3.4 Minimal polynomials and the first decomposition of a linear space

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- Annihilating and minimal polynomials of a linear transformation A at a subspace--properties

Theorem 3.25 Let $\mathcal{V}$ be linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\psi(\lambda) \in \mathcal{F}[\lambda]$ be the minimal polynomial of $\mathscr{A}$. Then

1. $m = \deg \psi(\lambda) \leq \dim \mathcal{V} = n$;

2. $\psi(\mathscr{A}) = \mathscr{O}$;

3. $\mathscr{A}^k$ is a linear combination of $\mathscr{E}$, $\mathscr{A}$, ..., $\mathscr{A}^{k-1}$ if and only if $k \geq m$.

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- Annihilating and minimal polynomials of a linear transformation A at a subspace--properties

Theorem 3.20 Let $\mathcal{V}$ be linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\mathcal{U} \subset \mathcal{W} \subset \mathcal{V}$ be subspaces of $\mathcal{V}$, $\alpha_{\mathcal{U}}(\lambda)$, $\beta_{\mathcal{W}}(\lambda)$ and $\gamma(\lambda)$ be the minimal polynomial of $\mathscr{A}$ at $\mathcal{U}$, $\mathcal{W}$ and $\mathcal{V}$ respectively. Then $\alpha_{\mathcal{U}}(\lambda)|\beta_{\mathcal{W}}(\lambda)$, $\beta_{\mathcal{W}}(\lambda)|\gamma(\lambda)$.

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- Relationship between the annihilating polynomial of a linear transformation A and that of the matrix representation A

Theorem 3.21.1 $\varphi_{\boldsymbol{\alpha}}(\lambda) \in \mathcal{F}[\lambda]$ is an annihilating polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$, if and only if it is an annihilating polynomial of the representation A of $\mathscr{A}$ at the coordinate $\boldsymbol{x}$ of $\boldsymbol{\alpha}$;

•Pg 117-118

Theorem 3.21.2 $\varphi_{\mathcal{W}}(\lambda)$ is an annihilating polynomial of $\mathscr{A}$ at the subspace $\mathcal{W} \subset \mathcal{V}$, if and only if $\varphi_{\mathcal{W}}(\lambda)$ is an annihilating polynomial of the representation A of $\mathscr{A}$ at the matrix W formed by the coordinates of the basis vectors of $\mathcal{W}$;

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Theorem 3.21.3 $\varphi(\lambda)$ is an annihilating polynomial of $\mathscr{A}$, if and only if $\varphi(\lambda)$ is an annihilating polynomial of the representation A of $\mathscr{A}$.

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- *3.4.2 The first decomposition of a linear space*

In the following, when a vector α or a subspace $\mathcal{W}$ is specified, the suffix in the associated annihilating or minimal polynomial will be abbreviated.

Theorem 3.22 Let $\mathcal{V}$ be a linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\mathcal{W} \subset \mathcal{V}$ be a subspace of $\mathcal{V}$ and $\mathscr{A}(\mathcal{W}) \subset \mathcal{W}$, $\psi(\lambda) \in \mathcal{F}[\lambda]$ be the minimal polynomial of $\mathscr{A}$ at $\mathcal{W}$ with the factorization

$$\psi(\lambda) = \varphi_1(\lambda)\varphi_2(\lambda)\cdots\varphi_s(\lambda)\,, \qquad (1)$$

where the polynomials $\varphi_1(\lambda)$, $\varphi_2(\lambda)$, ..., $\varphi_s(\lambda)$ are pairwise coprime, i.e.

$$\text{g.c.d}\left\{\varphi_j(\lambda)\,,\,\varphi_k(\lambda)\right\} = 1\,, \quad \forall j \neq k\,. \qquad (2)$$

Then $\mathcal{W}$ can be decomposed into the direct sum

$$\mathcal{W} = \mathcal{W}_1 \dot{+} \mathcal{W}_2 \dot{+} \cdots \dot{+} \mathcal{W}_s \qquad (3)$$

where the minimal polynomial of $\mathscr{A}$ at $\mathcal{W}_j$ is $\varphi_j(\lambda)$, and $\mathscr{A}(\mathcal{W}_j) \subset \mathcal{W}_j$, $j = 1, 2, \ldots, s$.

3.4 Minimal polynomials and the first decomposition of a linear space

Z. Song

- *Two important theorems useful in studying the structure of a linear space w.r.t a linear transformation*

Theorem 3.23 Let $\mathcal{V}$ be a linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\boldsymbol{\alpha}_j \in \mathcal{V}$, and $\varphi_j(\lambda)$ be the minimal polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}_j$, $j = 1, 2, \ldots, s$ with g.c.d$\left\{ \varphi_j(\lambda), \varphi_k(\lambda) \right\} = 1$ for all $j \neq k$. Then the minimal polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2 + \cdots + \boldsymbol{\alpha}_s$ is $\psi(\lambda) = \varphi_1(\lambda)\varphi_2(\lambda)\cdots\varphi_s(\lambda)$.

Theorem 3.24 Let $\mathcal{V}$ be linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\mathcal{W} \subset \mathcal{V}$ be $\mathscr{A}$-invariant, and $\psi(\lambda)$ be the minimal polynomial of $\mathscr{A}$ at $\mathcal{W}$. Then there exists $\boldsymbol{x} \in \mathcal{W}$ such that the minimal polynomial of $\mathscr{A}$ at $\boldsymbol{x}$ is $\psi(\lambda)$.

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- *The first decomposition of a linear space over the field C*

Let

$$[\hat{\boldsymbol{\alpha}}_1 \quad \hat{\boldsymbol{\alpha}}_2 \quad \dots \quad \hat{\boldsymbol{\alpha}}_n]) = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]X\,,$$

where the transition matrix $X \in \mathcal{C}_n^{n\times n}$ is such that $X^{-1}AX$ is in Jordan canonical form, then $\mathcal{V}$ can be decomposed into the direct sum of the Jordan subspaces as (3.56) and (3.57) on Pg 115.

In the following we shall study the minimal polynomial of $\mathscr{A}$ at the Jordan subspaces

$$\mathcal{W}_{ij} \triangleq \text{span}\left\{\hat{\boldsymbol{\alpha}}_{ij,1}\ ,\ \hat{\boldsymbol{\alpha}}_{ij,2}\ ,\ \dots\ ,\ \hat{\boldsymbol{\alpha}}_{ij,n_{ij}}\right\}.$$

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- *The first decomposition of a linear space over the field C*

From (3.46) and theorem 3.21 it is obvious that

$$
\begin{aligned}
& \lambda-\lambda_j \text{ is the minimal polynomial of } A \text{ at } \boldsymbol{x}_{ij,1} \\
\Longleftrightarrow\ & \lambda-\lambda_j \text{ is the minimal polynomial of } \mathscr{A} \text{ at } \widehat{\boldsymbol{\alpha}}_{ij,1}\,,
\end{aligned}
$$

$$
\begin{aligned}
& (\lambda-\lambda_j)^2 \text{ is the minimal polynomial of } A \text{ at } \boldsymbol{x}_{ij,2} \\
\Longleftrightarrow\ & (\lambda-\lambda_j)^2 \text{ is the minimal polynomial of } \mathscr{A} \text{ at } \widehat{\boldsymbol{\alpha}}_{ij,2}\,,
\end{aligned}
$$

$$
\begin{aligned}
& \vdots \\
& (\lambda-\lambda_j)^{n_{ij}-1} \text{ is the minimal polynomial of } A \text{ at } \boldsymbol{x}_{ij,n_{ij}-1} \\
\Longleftrightarrow\ & (\lambda-\lambda_j)^{n_{ij}-1} \text{ is the minimal polynomial of } \mathscr{A} \text{ at } \widehat{\boldsymbol{\alpha}}_{ij,n_{ij}-1}\,,
\end{aligned}
$$

$$
\begin{aligned}
& (\lambda-\lambda_j)^{n_{ij}} \text{ is the minimal polynomial of } A \text{ at } \boldsymbol{x}_{ij,n_{ij}} \\
\Longleftrightarrow\ & (\lambda-\lambda_j)^{n_{ij}} \text{ is the minimal polynomial of } \mathscr{A} \text{ at } \widehat{\boldsymbol{\alpha}}_{ij,n_{ij}}.
\end{aligned}
$$

3.4 Minimal polynomials and the first decomposition of a linear space

- *The first decomposition of a linear space over the field C*

Since $\hat{\boldsymbol{\alpha}}_{ij,1}$, $\hat{\boldsymbol{\alpha}}_{ij,2}$, ..., $\hat{\boldsymbol{\alpha}}_{ij,n_{ij}}$ form a basis for $\mathcal{W}_{ij}$, the last equation reveals that $(\lambda-\lambda_j)^{n_{ij}}$ is the minimal polynomial of $\mathscr{A}$ at $\mathcal{W}_{ij}$, and there exists $\boldsymbol{\alpha}_{ij} \in \mathcal{W}_{ij}$ such that the minimal polynomials of $\mathscr{A}$ at $\boldsymbol{\alpha}_{ij}$ and at the whole subspace $\mathcal{W}_{ij}$ are identical, see theorems 3.22 and 3.24.

On noting the factorization (3.32) of $\psi_i(\lambda)$, and g.c.d $\left\{ (\lambda-\lambda_j)^{n_{ij}}, (\lambda-\lambda_k)^{n_{ik}} \right\} = 1$ for all $j \neq k$, from theorem 3.22 it follows that $\psi_i(\lambda)$ is the minimal polynomial of $\mathscr{A}$ at the $\mathscr{A}$-invariant subspace

$$\mathcal{W}_i = \mathcal{W}_{i1} \dot{+} \mathcal{W}_{i2} \dot{+} \cdots \dot{+} \mathcal{W}_{is}\,, \qquad (1)$$

and

$$\mathcal{V} = \mathcal{W}_1 \dot{+} \mathcal{W}_1 \dot{+} \cdots \dot{+} \mathcal{W}_l\,. \qquad (2)$$

The minimal polynomial of $\mathscr{A}$ is thus the largest invariant polynomial $\psi_l(\lambda)$.

3.4 Minimal polynomials and the first decomposition of a linear space

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- Characterization annihilating and the minimal polynomials of a matrix A

Theorem 3.26 $\phi(\lambda)$ is an annihilating polynomial of A if and only if there exists a matrix polynomial $C(\lambda)$ such that

$$\phi(\lambda) \cdot E = C(\lambda)(\lambda E - A)$$

Theorem 3.27 Every square matrix A satisfies its own characteristic polynomial, i.e.

$$f_A(A) = A^n + a_{n-1}A^{n-1} + \cdots + a_1 A + a_0 E = 0$$

where $f_A(\lambda) = \det(\lambda E - A) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$.

Corollary 3.4 The minimal polynomial $m_A(\lambda)$ of the square matrix A is given by $m_A(\lambda) = \frac{f_A(\lambda)}{d(\lambda)}$.

3.4 Minimal polynomials and the first decomposition of a linear space

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- Implications
  - Note that similar matrices have the same annihilating and minimal polynomials,
  - Please see the last paragraph on Pg 123

- Generation to minimal polynomials of block matrices

Theorem 3.28 Let $A = \text{blockdiag}\{A_1, A_2, \ldots, A_l\}$, $\zeta_1(\lambda), \zeta_2(\lambda), \ldots, \zeta_l(\lambda)$ be the minimal polynomials of $A_1, A_2, \ldots, A_l$ respectively. Then the minimal polynomial of A is the least common multiplier of $\zeta_1(\lambda), \zeta_2(\lambda), \ldots, \zeta_l(\lambda)$.

- See example on Pg 124.

3.5 The cyclic invariant subspaces and the second decomposition of V

Z. Song

•3.5.1 The notion of cyclic invariant subspace

Let $\mathcal{V}$ be linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\boldsymbol{\alpha} \in \mathcal{V}$, then we can discuss the subspace spanned by the vectors

$$\boldsymbol{\alpha}, \quad \mathscr{A}(\boldsymbol{\alpha}), \quad \mathscr{A}^2(\boldsymbol{\alpha}), \quad \ldots, \quad \mathscr{A}^m(\boldsymbol{\alpha}), \quad \ldots$$

which can be understood as the linear space generated by the vector $\boldsymbol{\alpha}$ and the repeated actions of $\mathscr{A}$. From the minimal polynomial theory, for m large enough, $\mathscr{A}^{m+k}(\boldsymbol{\alpha})$, $k = 0, 1, \ldots$ can be represented by the first m vectors. Hence, we introduce the following definition.

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Z. Song

•3.5.1 The notion of cyclic invariant subspace

Definition 3.15 Let $\mathcal{V}$ be a linear space over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{V})$, $\boldsymbol{\alpha} \in \mathcal{V}$, m be the degree of the minimal polynomial of $\mathscr{A}$ at $\boldsymbol{\alpha}$

$$\mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha}) = \text{span}\left\{\boldsymbol{\alpha}, \quad \mathscr{A}(\boldsymbol{\alpha}), \quad \mathscr{A}^2(\boldsymbol{\alpha}), \quad \ldots, \quad \mathscr{A}^{m-1}(\boldsymbol{\alpha})\right\} \tag{1}$$

is called the cyclic subspace generated by the vector $\boldsymbol{\alpha}$ and the cyclical actions of $\mathscr{A}$. $\boldsymbol{\alpha}$ is called the generating element of $\mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha})$.

3.5 The cyclic invariant subspaces and the second decomposition of V

Z. Song

- 3.5.1 The properties of cyclic invariant subspace

Theorem 3.29 Let $\mathcal{F}$, $\mathcal{V}$, $\mathscr{A}$, $\boldsymbol{\alpha}$, m be defined in definition 3.15, then

1. $\dim \mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha}) = m$;
2. $\mathscr{A}[\mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha})] \subset \mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha})$;
3. for any $k \in \mathcal{Z}_{0,+}$, $\mathscr{A}^k(\boldsymbol{\alpha}) \in \mathcal{S}_{\mathscr{A}}(\boldsymbol{\alpha})$.

3.5 The cyclic invariant subspaces and the second decomposition of V

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•3.52 The second decomposition of V

We shall show the previous decompostion in the last section, showing below, is in fact decomposed V into the direct sum of a few cyclic subspaces.

$$\mathcal{W}_i = \mathcal{W}_{i1} \dot{+} \mathcal{W}_{i2} \dot{+} \cdots \dot{+} \mathcal{W}_{is}\,, \qquad (1)$$

and

$$\mathcal{V} = \mathcal{W}_1 \dot{+} \mathcal{W}_1 \dot{+} \cdots \dot{+} \mathcal{W}_l\,. \qquad (2)$$

Theorem 3.30 The $\mathscr{A}$-invariant subspace

$$\mathcal{W}_{ij} = \text{span}\left\{ [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_1 \quad \ldots \quad \boldsymbol{\alpha}_n] X_{ij} \right\}$$

is cyclic.

3.5 The cyclic invariant subspaces and the second decomposition of *V*

Z. Song

- 3.5.2 The second decomposition of *V*

  - Please refer to Pg 127-129, Wi is cyclic

  - This also gives the relation between N and J,

  - VN=JV, where V is the generalized Vandermonder matrix, please refer to Pg128.

  - End of decompositions of a space, all related to Jordan form, natural form, invariant polynomials, engivalues and engivectors (the way we look at matrices and linear transformation)

3.5 The cyclic invariant subspaces and the second decomposition of *V*

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